FIVUCSAS

Face and Identity Verification :: Cloud-based SaaS

> MULTI_TENANT_BIOMETRIC_AUTH.SYS -ACTIVE-LIVENESS -ANTI-SPOOF

> authors[] = {Ahmet Abdullah Gültekin, Ayşe Gülsüm Eren, Ayşenur Arıcı}
> advisor = Mustafa Ağaoğlu course = CSE4197 / CSE4198 term = Spring 2025-2026
> institution = Marmara University · Faculty of Engineering · Department of Computer Engineering

0%
[REPLAY_FAR]
n=120 recorded-video attempts

<1s
[E2E_LATENCY]
P95 = 950ms, CPU only

10
[AUTH_FACTORS]
face + voice + NFC + WebAuthn + ...

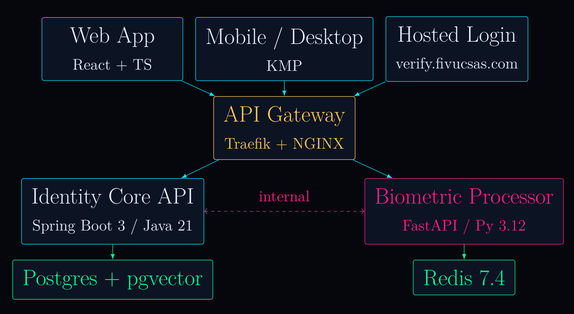
[01] PROBLEM

Online authentication is fragmented. Passwords leak, SMS OTPs are SIM-swappable, hardware keys exclude phone-only users, and most biometric services trust a **single passive signal** — a still selfie or a recorded clip — that a prepared attacker can replay.

GDPR / KVKK demand verifiable consent, the right to erasure, and data portability; consumer auth stacks treat these as an afterthought.

> **goal:** a tenant-configurable, multi-modal identity platform where any subset of **10** factors composes into MFA flows, and every biometric factor is protected by an *active*, randomised liveness challenge a recorded video cannot replay.

[02] SYSTEM TOPOLOGY



// biometric svc: no public route. docker net + api-key gated.

[03] PASSIVE || ACTIVE LIVENESS

[PASSIVE_ONLY]
single frame -> texture / colour cues.
✗ photo of phone screen *often passes*
✗ recorded clip *passes*
✗ deepfake injection *often passes*

[ACTIVE_CHALLENGE]
n-step facial-action sequence + landmark verification + passive layer.
✓ photo *fails*
✓ recorded clip *fails (sequence mismatch)*
✓ deepfake *must hit cues live*

[04] CONTRIBUTIONS

[C1] **Tenant-configurable composition** of 10 auth factors into MFA flows declared at admin layer.
[C2] **Active randomised liveness** — n-step facial-action sequence bound to per-attempt nonce, server-verified.
[C3] **Pipeline decomposition** along OIDC-contract / ML-iteration boundary into two services on a private Docker network.
[C4] **Embedding-at-rest encryption** — Facenet512 wrapped in Fernet (AES-128-CBC + HMAC-SHA-256) prior to pgvector write.
[C5] **Regulator-aware data lifecycle** — tenant-scoped soft-delete + scheduled purge + audit logs (GDPR Art. 15/17, KVKK).

[05] METHODS

Monolithic auth was rejected early: biometric ML changes weekly, but the OAuth / OIDC contract must remain frozen. The system is split along that boundary into two horizontally-scalable services on a private Docker network.

> per_verify_attempt():
[1] server issues randomised action sequence n=3-5 + nonce
[2] client streams 60-frame landmark vectors (MediaPipe FaceLandmarker, 468 pts)
[3] server checks each action vs EAR / MAR / yaw thresholds + temporal consistency
[4] UniFace MiniFASNet runs passive spoof check in parallel
[5] embedding (Facenet512) encrypted with Fernet (AES-128-CBC + HMAC-SHA-256), matched against user vector in pgvector (HNSW)

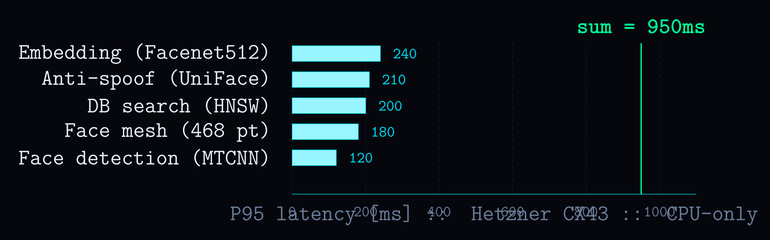
// 60 Flyway migrations · 5 git submodules · self-hosted GH runner on Hetzner CX43 · 1,820+ tests gating every PR · 8 ADRs.

[06] AUTH FACTORS

PASSWORD · EMAIL_OTP · SMS_OTP · TOTP · QR · FACE · VOICE · FINGERPRINT · WEBAUTHN · NFC_DOCUMENT (MRZ + chip)

// gold tiles = wrapped in active-liveness; any subset composable.

[07] EXPERIMENTS & RESULTS



quality_metric	target	observed
replay_attack_far (n=120)	< 1%	0.0%
passive_spoof_auc	> 0.95	0.97
backend_tests	green	633 / 633
web_vitest + playwright	green	619 + 27
mobile (KMP) units	green	425 / 425

[08] CONCLUSION

Production-grade, GDPR-compliant biometric auth *is* achievable on commodity hardware when the system is decomposed along its right axis (OIDC contract vs. ML model) and when liveness is active, not just passive. Measured 0 / 120 replay-attack false-accepts and a 0.97 passive-spoof AUC are both within practical deployment limits on a single CPU server.

[09] FUTURE_WORK[]

> **BYOD tenant DB.** Enterprise tenants host biometric data in their own database, schema-pinned.
> **On-device pre-filter.** Stripped MediaPipe drops obvious non-faces before upload.
> **Cross-modal challenge.** Voice nonce + facial action to defeat audio-only deepfakes.
> **Verifiable credentials.** W3C VCs after successful KYC for downstream portability.

[10] DEMO :: REF

scan to test live demo
verify.fivucsas.com

// refs
[1] Serengil, S., Özpınar, A. *LightFace / DeepFace.* IEEE ASYU, 2020.
[2] Lugaresi, C. et al. *MediaPipe.* arXiv:1906.08172.
[3] Soukupová, T., Čech, J. *Real-Time Eye Blink Detection.* CVWW, 2016.
[4] Wan, L. et al. *Generalized End-to-End Loss for Speaker Verification.* ICASSP, 2018.
[5] Yadav, S. et al. *UniFace.* CVPR-W, 2023.

» github.com/Rollingcat-Software/FIVUCSAS :: fivucsas.com :: ahmetabdullahgultekin@gmail.com :: Marmara University 2026